Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 10 September 2026 Time: Not recorded
Group members present	Rohan Mathew, Jianhao Zhang, Xuan Zhang, Ria Mishra, Xingcan Wang (Alex). Supervisor: Saumya.
Apologies	None

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Present progress from the previous week and the current week.	All members	This meeting
Develop physics based signal analysis and change point detection approaches.	Team	Not recorded

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Physics conditioned wavelets	Rohan described applying wavelets to a derived physical channel, such as apparent viscosity from stress/shear rate. A preliminary median timing gap reportedly reduced from about 11 to 6 seconds; this was not proof of a physical gel point.
Limits of clotting inference	Driven step/LAOS tests and preformed clot data do not necessarily contain clot formation. The exploratory supervised model lacked suitable ground truth and did not provide reliable clotting evidence.
Multivariate change point framework	Xuan outlined preprocessing, rheological feature construction, PELT or binary segmentation, and persistence/physics filters. Storage/loss moduli, loss tangent, viscosity, gradients and rolling statistics were discussed.
Operational transition time	Jianhao presented a reproducible PELT workflow across rheology, TEG and digitized ROTEM curves. The result was described as an operational transition time; differing biological reference stages can produce systematic timing differences.
Robustness and further methods	Sensitivity checks were encouraging. Saumya requested comparison with two or three additional change point algorithms. Conformal prediction was suggested as an optional uncertainty approach if time permitted.
External clotting data comparison	Alex compared STFT and wavelet methods on human plasma stiffness curves. Classification and timing favoured different methods, with noise sensitivity and transfer to whole blood remaining limitations.
Project consolidation	Focus on which methods suit which data properties rather than declaring one universal winner. Incorporate client feedback, organize existing techniques and consolidate the team work into one pipeline and report.

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3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Compare PELT with two or three additional change point algorithms.	Jianhao	Not recorded
Investigate noise sensitivity and test suitable additional real clotting data.	Xingcan (Alex)	Not recorded
Continue physically meaningful wavelet analysis and document its limitations.	Rohan	Not recorded
Consider uncertainty quantification if the schedule permits.	Jianhao	Optional; time permitting
Present remaining work to the client, consolidate feedback and agree a shared pipeline.	All members	After client feedback
Share client feedback with Saumya and organize methods/results for the report.	All members	Before next meeting

4 Next meeting

Location	Not recorded
Date & Time	Next week; exact date and time not recorded
Goals	Begin with client feedback, review method comparisons and agree the consolidated analysis direction.

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